EXPERIMENT NO: 8

Conduct Z test for the Data Given using Python Objective

PROGRAM:

```
import numpy as np
import scipy.stats as stats
# Define the sample data (hypothetical weights in grams)
sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152,
151,148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])
# Population mean under the null hypothesis
population_mean = 150
# Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation
# Number of observations
n = len(sample_data)
# Calculate the Z-statistic
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
# Calculate the p-value
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test
# Print results
print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")
# Decision based on the significance level
alpha = 0.05
if p_value < alpha:
print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")
else:
print("Fail to reject the null hypothesis: There is no significant difference in average weight from
150 grams.")
```

OUTPUT:

```
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from 150
grams.
```